

Application of the PPP Method to the Calculation of Ionization Potentials and Electron Affinities of Conjugated Organic Molecules

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Abstract

By making the atomic effective charge self-consistent with the molecular charge density, it is shown that the PPP method can accurately predict the ionization potential and electron affinity computed as an energy difference of separately minimized ground and ionized states. The calculation is applied to a large variety of conjugated organic molecules, including heterocyclic systems and π -electron-contributing substituent groups.

I. Background

It is well known that calculation of the ionization potential (IP) from Koopman's theorem applied to ground state wave functions obtained with the Pariser-Paar-Pople method [2] can yield differences of 1–1.5 eV from the experimental value. In the years 1960–1962, shortly after Roothaan [3] developed an open-shell LCAO-MO method, Hoyland and Goodman [4] analyzed the reasons for this failure and proposed four essential modifications to be included in the PPP method for successful IP and EA calculations. These were: (1) calculate the IP or EA as an energy difference of a closed-shell singlet state (usually the ground state) and an open-shell doublet state (usually the ion); (2) adjust the atomic valence state ionization potential in the ion; (3) correct the one-center Coulomb integral in the ion; and (4) allow for sigma electron polarization during the ionization process. Note that even for calculating the vertical IP, which corresponds to a "fast" measurement where the nuclei retain the ground state configuration in the ion, the rearrangement of all the electrons must be accounted for. For example, Hoyland and Goodman show that the contribution of the σ term (4) is ~ 0.8 eV for a polyacene.

The PPP method so modified retains enough physics to become a powerful analytical tool; for example, in 1970 Nelson and Selsby [4] applied this theory to probe the interaction of photosensitizing cyanine dyes adsorbed on semiconducting substrates.

In 1977, Selsby et al. [6] proposed an automatic method of incorporating the Hoyland and Goodman corrections into the PPP method. The procedure is based on making the local atomic effective nuclear charge, Z_i , a function of the molecular π charge density. All atomic semiempirical parameters are expressed in terms of Z , and

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a computational scheme is employed in which the effective charge becomes self-consistent with the calculated molecular charge density for each state considered. By varying the nearest-neighboring bond length in terms of Z , the adiabatic as well as the vertical IPs and EAS were obtained. In their recent book *Electronic Processes in Organic Crystals*, Pope and Swenberg [7] refer to this variable Z calculation as an accurate method of obtaining vacuum IPs and EAS, of considerable value to the organic solid state scientist.

II. Modification and Additions to the Variable Z PPP Method

It is clear on electrostatic grounds that each of the four conditions required by Hoyland and Goodman are achieved through the effective charge variation. For example, as Z_i increases in the positive ion, due to the loss of π charge at each atomic site, the valence state ionization potential, W_i , becomes more negative, while the one-center integrals, become more positive. Thus, with the variable Z approach, computing the IP or EA as an energy difference not only allows for electronic rearrangement in the ion but also adjusts all the atomic semiempirical parameters in the correct sense, consistent with the physical process taking place. The σ framework correction is accounted for by proposing a σ electron adjustment term in the formula for Z_i which is essentially proportional to the change in π charge density at the atomic site. This σ electron contraction term automatically dampens the changes in W and γ by the amounts proposed by Hoyland and Goodman for the σ polarization correction.

The variable Z PPP method produced excellent results on the conjugated molecules tried. However, the method required the use of benzene and naphthalene experimental IPs to determine the two "molecular parameters" used to calculate the sigma contraction from the change in π electron density.

In this work we simplify and analyze the basic formula for the effective charge, reducing the number of "molecular parameters" to one. In Appendix A we show that this parameter can be derived to a close approximation. In addition, the variable Z PPP method is extended to include sulfur heteroatoms and to include the π electron contributing functional groups; $-\text{NH}_2$, $-\text{OH}$, $-\text{NO}_2$, and $-\text{COOH}$. This greatly increases the utility of the calculation.

We note here that the nearest-neighbor R_{ij} variation, used to calculate adiabatic IPs and EAS in [6], has been omitted to shorten this work. In general, the adiabatic IPs and EAS differed by 0.15 eV or less from the vertical values, and the calculation can be included if a particular interest arises.

III. Outline of the Variable Z PPP Method

The flowchart for the method is shown in Figure 1. The central feature of the scheme is the formula for the effective charge at the i th atomic site, Z_i , which is given by

$$Z_i = Z_{\text{Slater}} - 0.35 \times (Q_i - \alpha_i) + k \times (Q_i - Q_i^s) \quad (1)$$

where Z_{Slater} is the nominal value computed from Slater's rules [8], Q_i is the π charge

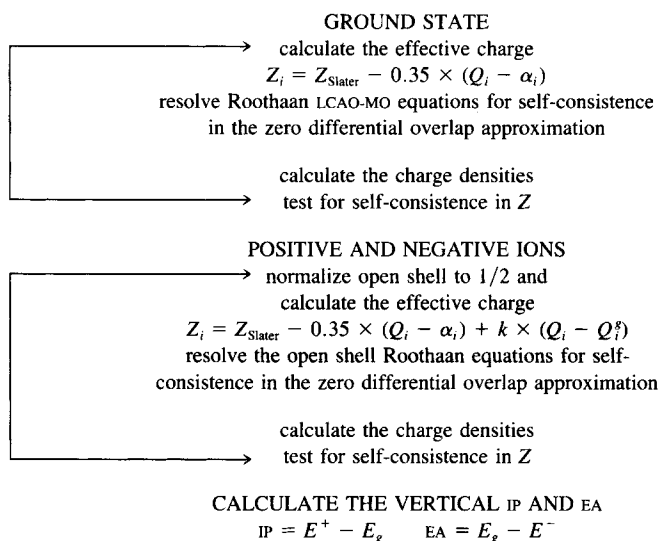


Fig. 1. Flowchart for the variable Z PPP method for vertical IPs and EAs.

density at the site, α_i is the nominal number of π electrons contributed by the atom at the site, and Q_i^g is the ground state charge density. The first correcting term allows Z to adjust to the particular electronic molecular charge density through the difference between the calculated π charge and the nominal π charge at the site. The second correcting term occurs only in the calculation of the doublet state and represents the effect of the sigma contraction upon gain or loss of π charge.

Physical arguments leading to the form of Eq. (1) and a derivation of an "upper bound" of 0.339 for the constant k are presented in Appendix A. However, for the purpose of obtaining the optimum numerical results we must retain one "molecular parameter" by setting $k = 0.256$, which yields the experimental IP of naphthalene. The σ contraction in the formation of the anion is less than that of the cation. In accordance with the previous work [6] we reduce k for the anion by a factor of $(0.81/0.91)^2$, which is the square of the ratio of π electron "heights" in a six and seven electron carbon atom, respectively.

The atomic semiempirical parameters as functions of the effective charge Z are the same as in the previous work [6] and are summarized in Appendix B.

IV. Results and Discussion

The vertical IPs and EAs reported here are calculated using standard bond lengths and angles for the ring structures and experimental data for the bond length and angles of the functional groups —OH, —NH₂, —NO₂, and —COOH. We also note that placement of these functional groups in the *meta* position of a single ring, or use of more than two functional groups on a single ring, produces oscillations in the self-consistent calculation and requires a density averaging technique to achieve convergence.

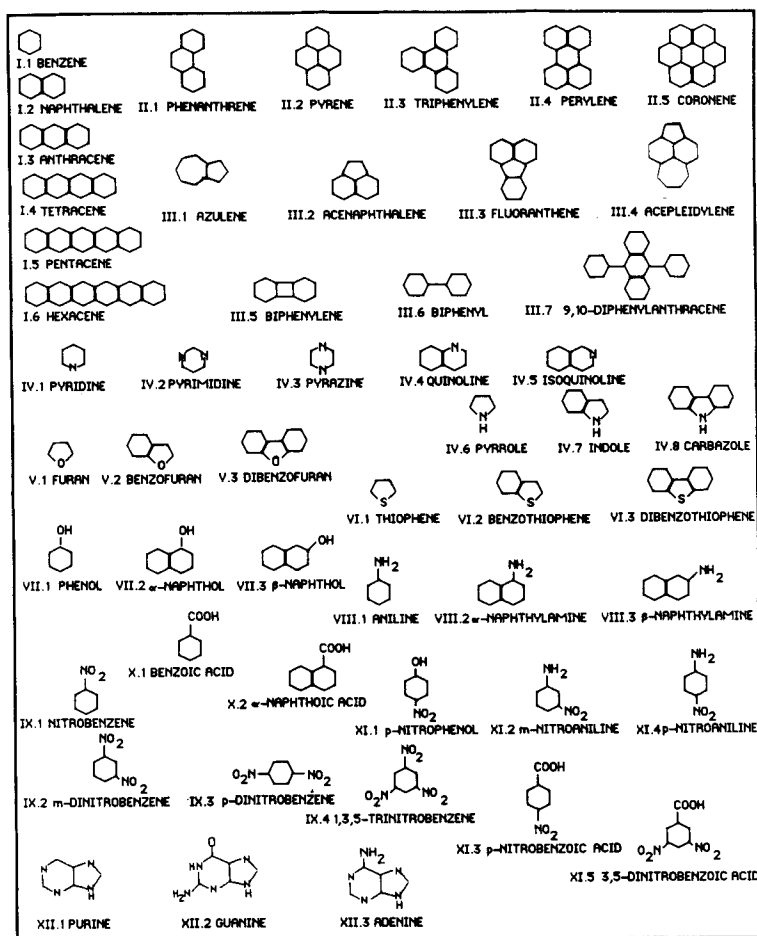


Fig. 2. Molecules studied in this work. Roman numbering refers to Table I.

Figure 2 shows the molecules considered. The calculated vertical IPs and EAS are compared with experimental values in Table I. High resolution photoelectron spectra (PS) for the molecules types I through VI generally yield unambiguous peaks corresponding to the lowest IP. For example, for the polyacenes, the PS of Clark et al. [21] are in agreement with those used in Table I, of Boschi et al. [8], to within 0.1 eV. Even when the peaks are broader, such as with pyrazine (IV.3), the PS of Eland [10, 11] and Baker and Turner [12] differ by only 0.03 eV. The experimental situation for the electron affinities is more complicated and is detailed in our earlier paper [6]. Table I lists experimental values obtained by Belkind and Grechov [9], Nenner and Schultz [13], as well as estimates made from [6].

Our predicted IPs and EAS are generally accurate for almost all of the IPs and EAS considered, falling within 0.2 eV of the experimental values. One interesting result is the prediction that addition of the NO_2 group stabilizes the anion.

TABLE I. Vertical ionization potential and electron affinity (in eV).

mol	IPcal	EAcad	IPexp	EAexp	mol	IPcal	EAcad	IPexp	EAexp
Polyacenes					Nitrogen Heteroatoms				
I.1	9.44	-1.53	9.25 ^a	-1.38 ^b	IV.1	9.67	-1.38	9.60 ^c	-0.62 ⁱ
I.2	—	-0.35	8.15 ^a	-0.28 ^b	IV.2	9.90	-0.77	9.90 ^c	
I.3	7.38	0.35	7.47 ^a	0.46 ^b	IV.3	9.91	-1.18	9.65 ^c	
I.4	6.87	0.78	7.04 ^a	0.97 ^b	IV.4	8.38	0.06	8.67 ^c	
I.5	6.57	1.05	6.74 ^a	1.32 ^b	IV.5	8.34	0.07	8.53 ^c	
I.6	6.35	1.24	6.44 ^a		IV.6	8.35	-2.87	8.20 ^c	
					IV.7	7.94	-1.36	7.90 ^d	
					IV.8	7.99	-1.01		
Condensed Ring					Oxygen Heteroatoms				
II.1	7.93	-0.22	7.86 ^b	0.03 ^g	V.1	8.68	-2.43	8.90 ^d	
II.2	7.37	0.30	7.41 ^b	0.38 ^g	V.2	8.19	-1.24	8.36 ^d	
II.3	8.04	-0.42	8.08 ^b		V.3	8.23	-0.91	8.40 ^d	
II.4	6.99	0.61	7.00 ^a	0.83 ^g	Sulfur Heteroatoms				
II.5	7.47	0.04	7.60 ^b		VI.1	8.93	-1.11	8.87 ^h	
Nonbenzenoid Aromatics					VI.2	8.55	-0.82		
III.1	7.15	0.70	7.43 ^a	0.86 ^g	VI.3	8.38	-0.73		
III.2	7.85	0.71	8.22 ^a	0.85 ^g					
III.3	7.91	0.53	7.95 ^a	0.73 ^g					
III.4	7.19	0.55	7.13 ^a						
III.5	7.85	-0.12	7.61 ^a						
III.6	8.41	-0.64	8.35 ^c	-0.21 ^g					
III.7	7.06	0.53	7.10 ^a						
SUBSTITUTED BENZENES AND NAPHTHALENES									
—OH Group					—COOH Group				
VII.1	8.61	-1.97	8.70 ⁱ		X.1	9.63	0.20	9.73 ^k	
VII.2	7.73	-0.61			X.2	8.34	0.67		
VII.3	7.81	-0.58	7.83 ^b		Mixed Groups				
—NH ₂ Group					XI.1	9.38	2.07	9.52 ⁱ	
VIII.1	8.11	-2.14	8.00 ⁱ		XI.2	8.77	2.09	8.80 ^h	
VIII.2	7.38	-0.75	7.30 ^b		XI.3	10.28	2.58		
VIII.3	7.53	-0.68	7.50 ^b		XI.4	8.90	1.95	8.85 ^h	
—NO ₂ Group					XI.5	10.72	3.07		
IX.1	9.99	2.31	9.92 ⁱ		Bases				
IX.2	10.74	2.91			XII.1	8.78	-0.61		
IX.3	10.74	3.08			XII.2	7.56	-1.06		
IX.4	11.31	3.28			XII.3	7.81	-1.39		

Note: The numbering of the molecules refers to Figure 2. For references, a = 8; b = 9; c = 10; d = 11; e = 12; f = 13; g = 14; h = 15; i = 16; j = 17; k = 18.

V. Summary

The original intent of Pariser, Parr, and Pople was to develop an easily applicable semiempirical theory that retains sufficient quantum physics to have molecular predictability. The variable Z feature, which incorporates the Hoyland-Goodman corrections in a simple algorithm, allows the PPP method to be applicable to ions with equal facility.

Appendix A: Formulation of the Zeta Density EQ. (1)

According to Slater's rules, the effective charge of a π atomic orbital is given by

$$Z_i = N_i - 2 \times v1 - \sigma \times v2 \quad (\text{A.1})$$

where N_i is the nuclear charge, $v1 = 0.85$ for the two $1s$ electrons, and $v2 = 0.35$ for the σ electrons. The natural extension of Eq. (A.1) which allows Z to vary according to the local π density fluctuations from the nominal number of π electrons contributed, α_i , is given by

$$Z_i = N_i - 2 \times v1 - (Q_i - \alpha_i) \times v2 - \sigma \times v2 \quad (\text{A.2})$$

where Q_i is the molecular π charge density at the site. Thus, if for a carbon site, $Q_i < 1.0$, Z increases by the amount $-(Q_i - 1) \times 0.35$. However, we must also account for the σ contraction in the ion. (As π charge is lost, the σ charge moves closer to the nucleus.) To account for this we rewrite Eq. (A.2) in the form

$$Z_i = N_i - 2 \times 0.85 - (Q_i - \alpha_i) \times 0.35 - \sigma \times v(r) \quad (\text{A.3})$$

where $v(r)$ is a radially dependent screening factor. To determine $v(r)$ we use the relationship of radii and effective charge for a hydrogenic atom,

$$\langle r \rangle = \frac{n^2 a_0}{Z} \left[1 + \frac{1}{2} \left(1 - \frac{L(L+1)}{n^2} \right) \right] \quad (\text{A.4})$$

where $a_0 = 0.529$ angstroms. For carbon $\langle r2p \rangle = 0.814 \text{ \AA}$ and $\langle r1s \rangle = 0.139$. From Slater's rules, $v(0) = 1.0$, $v(\langle r1s \rangle) = 0.85$ and $v(\langle r2p \rangle) = 0.35$, we may approximate $v(r)$ by an exponential of the form $\exp(-1.29r)$. Taking the Taylor expansion of this exponential about $r = \langle r2p \rangle$, we obtain

$$\begin{aligned} v(r) &= v(\langle r2p \rangle) + \frac{dv}{dr}(\langle r2p \rangle) \times (r - \langle r2p \rangle) \\ &= 0.35 - 0.451 \times (r - \langle r2p \rangle) \end{aligned} \quad (\text{A.5})$$

Since the core adjustment is a function of Z

$$r = \langle r2p \rangle + (dr/dZ) dZ, \quad (\text{A.6})$$

so that (A.5) becomes

$$v(r) = 0.35 - 0.451 \times (dr/dZ) \times dZ. \quad (\text{A.7})$$

From (A.4), $(dr/dZ) = -5a_0/Z^2 = -0.250$, so that

$$v(r) = 0.35 + 0.113 \times dZ \quad (\text{A.8})$$

where dZ is the change in Z due to the σ electrons.

Defining

$$d\pi = (Q_i - Q_i^q) \quad (\text{A.9})$$

we note that

$$dZ = -b \times d\pi \quad \text{with } b < 1 \quad (\text{A.10})$$

Equation (A.10) follows from the fact that dZ is induced by $d\pi$ so that $|dZ| < |d\pi|$. Thus, the proportionality constant b is less than 1. Further, a decrease in π charge produces an increase in Z , implying the minus sign. Thus,

$$v(r) = 0.35 - b \times 0.113 \times (Q_i - Q_i^g) \quad (\text{A.11})$$

Substituting (A.11) into (A.3) with $\sigma = 3$ and noting that $Z_{\text{Slater}} = N - 2 \times 0.85 - \sigma \times 0.35$, we obtain

$$Z_i = Z_{\text{Slater}} - 0.35 \times (Q_i - \alpha_i) + b \times 0.339(Q_i - Q_i^g) \quad (\text{A.12})$$

Equation (1) with the empirical parameter is

$$Z_i = Z_{\text{Slater}} - 0.35 \times (Q_i - \alpha_i) + 0.255(Q_i - Q_i^g) \quad (1)$$

Indeed, comparison of Eq. (A.12) with the empirical equation (1) confirms our interpretation of the two correcting factors to the Slater value of Z . We find a term due to the variance of the π charge at the site from the nominal number, corrected by a counteracting σ contraction induced by the change in π charge from the ground state value.

Appendix B: The Atomic Semiempirical Parameters

The valence state ionization potentials and one-center Coulomb integrals were obtained from spectroscopic data as described in Ref. 6.

$$W_i = aZ_i^2 + bZ_i + c \quad (\text{B.1})$$

$$\gamma_i = gZ_i \quad (\text{B.2})$$

The nearest-neighboring core integrals are given by the Mullikin formula [19]

$$\beta_{rs} = 0.53(W_r + W_s) \times S_{rs}/(1 + S_{rs}) \quad (\text{B.3})$$

where $\beta_{cc} = -2.39$ (eV) for $R_{cc} = 1.39$ Å.

The two-center Coulomb integrals are evaluated according to Mataga and Nishimoto [20] by

$$\gamma_{rs} = 14.397/[28.794/(\gamma_{rr} + \gamma_{ss}) + R_{rs}] \quad (\text{B.4})$$

TABLE II. Constants for Eqs. (B.1) and (B.2) and nominal values for W and γ attained when $Z = Z_{\text{Slater}}$.

At.	N	σ	α	Z_{Slater}	a	b	c	W (eV)	g	γ (eV)
C ⁺	6	3	1	3.25	-3.42	8.34	-2.52	-11.54	3.41	11.08
N ⁺	7	4	1	3.90	-1.15	-4.19	19.71	-14.12	3.16	12.33
N ⁺²	7	3	2	4.25	-1.30	-2.53	5.39	-28.85	3.90	16.58
O ⁺	8	5	1	4.55	-0.87	0.00	0.00	-17.91	2.87	13.06
O ⁺²	8	4	2	4.90	-0.48	-12.78	39.92	-34.21	3.89	19.06
S ⁺²		4	2	5.80	-0.33	-6.291	20.87	-23.96	1.95	11.32

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